Object-Oriented Paradigm Review

Abstract Data Type
Object-Oriented Paradigm
Inheritance
Polymorphism
Method Overloading
Method Overriding
What is an Interface
Polymorphism Examples
Class Relationships and Change Propagations

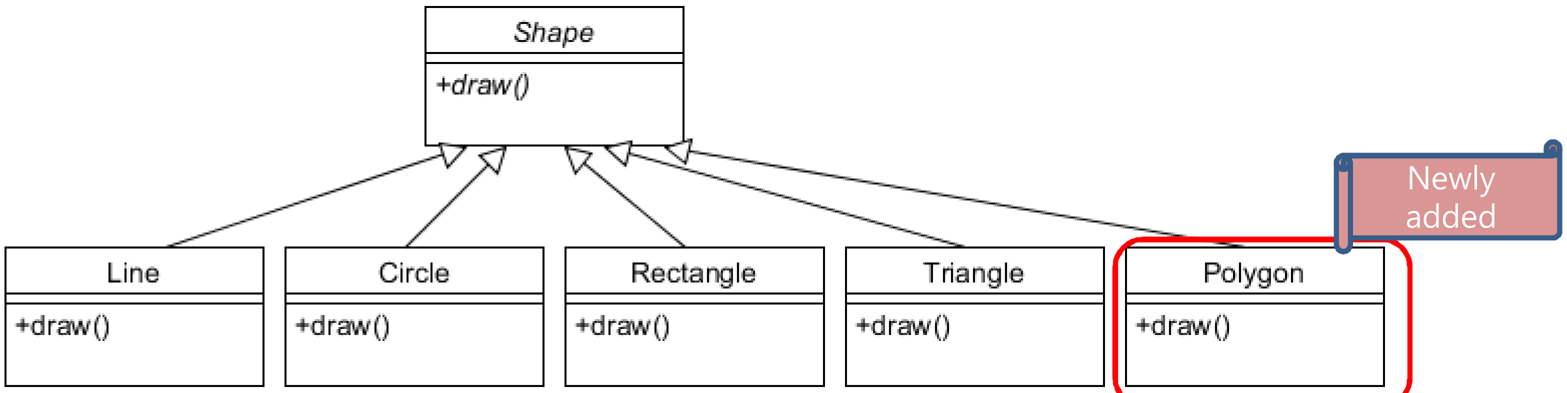
Data Abstraction

- Abstract Data Type (ADT)
  - **Encapsulation** of data and related operations into a **single syntactic unit**
  - Single syntactic unit 하나의 문법단위
    - increases program **organization**, **modifiability** (everything associated with a data structure is together), and **separate compilation**
  - Encapsulation 캡슐화.
    - increases **reliability;** by hiding the data representations, user code cannot directly access objects of the type or depend on the representation, **allowing the representation to be changed without affecting user code**

Object-Oriented Paradigm

Class = ADT + Inheritance + Polymorphism

Promotes reusability and flexibility



부모 코드의 재사용

Object-Oriented Paradigm

Class = ADT + Inheritance + Polymorphism

```
class ClientOfShape
{
   void renderShape(Shape s)
   {
      if (s.type == LINE) DrawLine(s.data);
      else if ((s.type == CIRCLE) DrawCirlce(s.data);
      else if ((s.type == RECT) DrawRect(s.data);
      else if ((s.type == TRIANGLE) DrawTriangle(s.data);
   }
}
```

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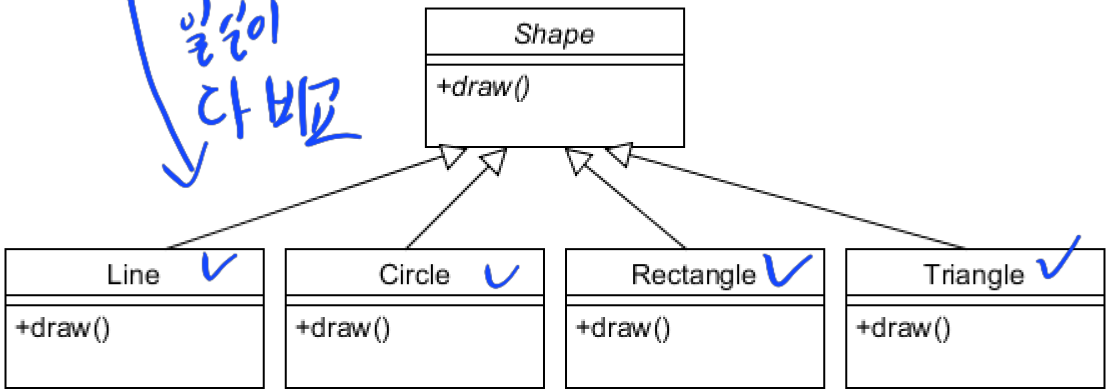
다 비교

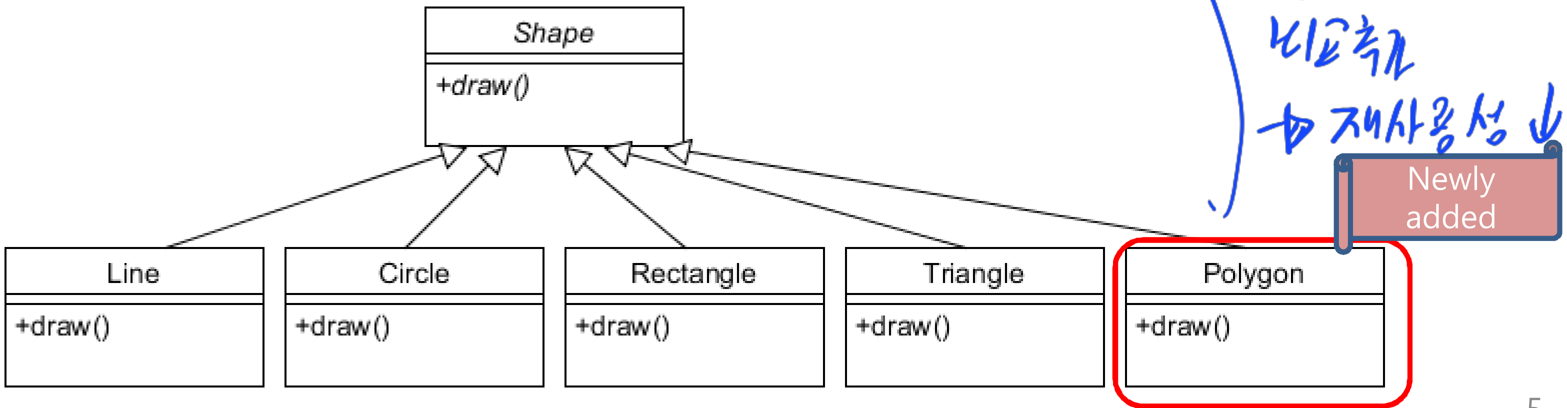


Object-Oriented Paradigm

Class = ADT + Inheritance + Polymorphism

```
class ClientOfShape
{
    void renderShape(Shape s)
    {
        if (s.type == LINE) DrawLine(s.data);
        else if ((s.type == CIRCLE) DrawCirlce(s.data);
        else if ((s.type == RECT) DrawRect(s.data);
        else if ((s.type == TRIANGLE) DrawTriangle(s.data);
        else if ((s.type == POLYGON) DrawPolygon(s.data);
    }
}
```

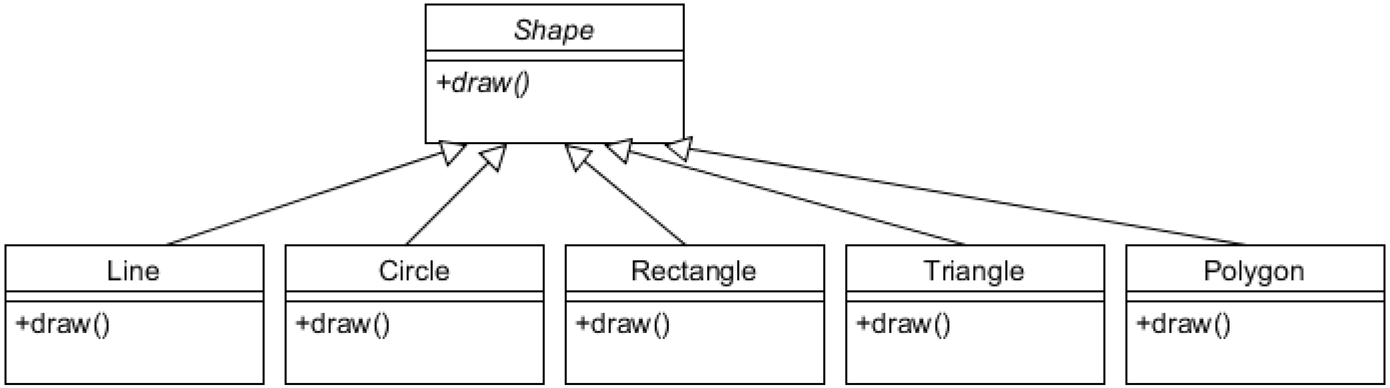
얘에 대한
비교추가
→ 재사용성 ↓



Object-Oriented Paradigm

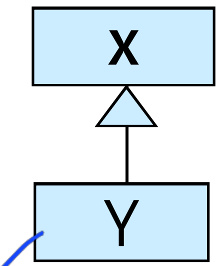
Class = ADT + Inheritance + Polymorphism

```
class ClientOfShape
{
    void renderShape(Shape s)
    {
        s.draw(); // No Change!
    }
}
```



Inheritance

- When class Y inherits from class X
  - Y inherits all the methods of X
    - Instances of Y can be sent all the messages that X responds to
  - Y inherits all the data from X
    - Instances Y have instances of all the data of X
  - As a consequence we can say
    - Y is a X → X is a Y (false)
  - Every instance of Y is also an instance of X
  - **We can use the instance of Y wherever the instance of type X is expected**



Y는 X가 하던걸 모든 handling 가능.

↳ X를 넣어야 하는 자리에 Y를 넣어도 문제가 없다.

Polymorphism

- A property of object oriented software by which an operation (typically virtual) may be performed in different ways in different classes.
  - Requires that there be multiple methods of the same name
  - The choice of which one to execute depends on the object that is in a variable
  - Reduces the need for programmers to code many if-else or switch statements
- Types of Polymorphism
  - Runtime polymorphism (Dynamic polymorphism)
    - Method Overriding
  - Compile time polymorphism (Static polymorphism)
    - Method Overloading

Method Overloading → Compile 할때 알수있음.

```
class X {
   void methodA(int num) {
        System.out.println ("methodA:" + num);
   }
   void methodA(int num1, int num2) {
        System.out.println ("methodA:" + num1 + "," + num2);
   }
   double methodA(double num) {
        System.out.println("methodA:" + num);
        return num;
   }
}

public class test {
   public static void main (String args []) {
        X Obj = new X();
        double result;
        Obj.methodA(20);
        Obj.methodA(20, 30);
        result = Obj.methodA(5.5);
        System.out.println("Answer is:" + result);
   }
}
```

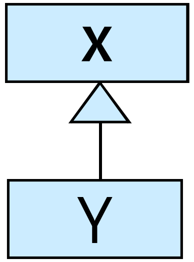
```
methodA:20
methodA:20,30
methodA:5.5
Answer is:5.5
```

Method Overriding

```
public class X {
    public void methodA() { //Base class method
        System.out.println ("hello, I'm methodA of class X");
    }
}

public class Y extends X {
    public void methodA() { //Derived Class method
        System.out.println ("hello, I'm methodA of class Y");
    }
}

public class Z {
   public static void main (String args []) {
       X obj1 = new X(); // Reference and object X
       X obj2 = new Y(); // X reference but Y object
       obj1.methodA();
       obj2.methodA();
   }
}
```

자식 포인터로 부모 가리킴 O

↳ 역은 X




```
hello, I'm methodA of class X
hello, I'm methodA of class Y
```

(Run-time) Polymorphism

- A *polymorphic variable* can be defined in a class that is able to **reference** (or point to) **objects of the class** and **objects of any of its descendants** (자기자신 + 자기의 자손)

- When a class hierarchy includes classes that override methods and such methods are called through a polymorphic variable, the binding to the **correct method** will be dynamic

- Allows software systems to be **more easily extended during both development and maintenance**

Abstract Class and Abstract Method (in Java)

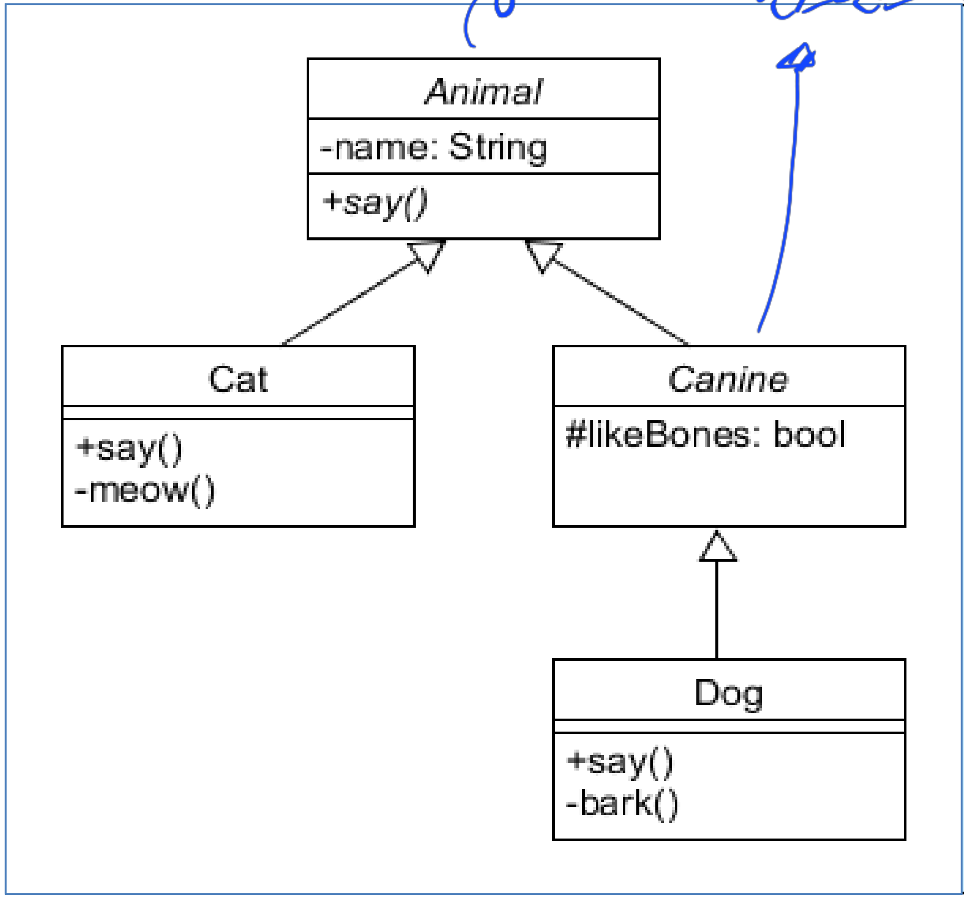
- An *abstract method* is one that does not include a definition (it only defines a protocol)
- A class defined with “abstract” keyword is an abstract class
- A class with at least one abstract method should become an abstract class (but not vice versa) abstract method 포함 클래스라면 반드시 abstract class다.
- An abstract class cannot be instantiated
- But, a variable of abstract class type can refer to any objects of its decedents!

Polymorphism + Abstract class

<pre>abstract class Animal {
    protected String name;
    abstract public void say();
}
 	
class Cat extends Animal {
    private void meow() { ... }
    public void say() { meow(); }
}
 	
abstract class Canine extends Animal {
    protected bool likeBones;
}
 	
class Dog extends Canine {
    private void bark() { ... }
    public void say() { bark(); }
}</pre>
	<pre>Animal a = null;
	Dog baduki = new Dog();
	Cat nabi = new Cat();
	
	a = baduki; a.say();
	a = nabi; a.say();
	
	Animal c1 = new Animal();
	Animal c2 = new Cat();
	Cat c3 = new Animal();
	Cat c4 = new Cat();
	
	Animal d1 = new Canine();
	Animal d2 = new Dog();
	Canine d3 = new Dog();
	Canine d4 = new Cat();
	Dog d5 = new Canine();
	Dog d6 = new Dog();</pre>

Polymorphism + Abstract class

abstract Class는 인스턴스 생성 불가능.




```
Animal  a = null; O
Dog  baduki = new Dog(); O
Cat  nabi = new Cat(); O
                a의 자손.
a = baduki; a.say(); O → bark()
a = nabi;  a.say(); O → meow()

Animal c1 = new Animal(); X
Animal c2 = new Cat(); O
Cat c3 = new Animal(); X
Cat c4 = new Cat(); O

Animal d1 = new Canine(); X
Animal d2 = new Dog(); O
Canine d3 = new Dog(); O
Canine d4 = new Cat(); X
Dog d5 = new Canine(); X
Dog d6 = new Dog(); O
```

Polymorphism + Abstract class

```
abstract class Animal {
    protected String name;
    abstract public void say();
}

class Cat extends Animal {
    private void meow() { ... }
    public void say() { meow(); }
}

abstract class Canine extends Animal{
    protected bool likeBones;
}

class Dog extends Canine {
    private void bark() { ... }
    public void say() { bark(); }
}
```

```
Animal  a = null;
Dog   baduki = new Dog();
Cat   nabi = new Cat();

a = baduki; a.say();
a = nabi;   a.say();

Animal c1 = new Animal(); //Compile Error!
Animal c2 = new Cat();
Cat c3 = new Animal(); //Compile Error!
Cat c4 = new Cat();

Animal d1 = new Canine(); //Compile Error!
Animal d2 = new Dog();
Canine d3 = new Dog();
Canine d4 = new Cat(); //Compile Error!
Dog d5 = new Canine(); //Compile Error!
Dog d6 = new Dog();
```

How a decision is made about which method to run

가장 밑에서 위로 올라가면서 method 찾기.

Step 1: If there is a concrete method for the operation in the current class, run that method.

Step 2: Otherwise, check in the immediate superclass to see if there is a method there; if so, run it.

Step 3: Repeat step 2, looking in successively higher superclasses until a concrete method is found and run.

Step 4: If no method is found, then there must be an error

- In Java and C++ the program would not have compiled

What is an Interface?

- An interface is similar to an abstract class with the following exceptions:
  - All methods defined in an interface are abstract; Interfaces cannot contain any implementation
  - Interfaces cannot contain instance variables.
  - However, they can contain public static final variables (i.e. constant class variables)
- Interfaces are declared using the "interface" keyword
  - If an interface is public, it must be contained in a file which has the same name.
- Interfaces are more abstract than abstract classes
- Interfaces are implemented by classes using the "implements" keyword.

Declaring an Interface

extend는 한 번만,
implement는 여러번 가능.

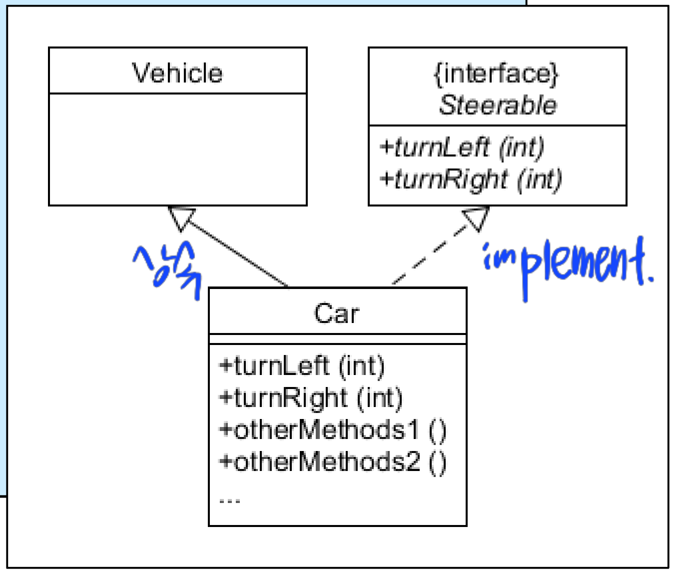
In Steerable.java:

```
public interface Steerable
{
  public void turnLeft(int degrees);
  public void turnRight(int degrees);
}
```

When a class "implements" an interface, the compiler ensures that it provides an implementation for all methods defined within the interface.

In Car.java:

```
public class Car extends Vehicle implements Steerable
{
  public void turnLeft(int degrees)
  {    …              }
  public void turnRight(int degrees)
  {    ...                   }
  public otherMethods1()  { ... }
  public otherMethods2()  { ... }
      …
 }
```



Interfaces as Types (Important !)

- When a class is defined, the compiler views the class as a new type.

- The same thing is true of interfaces. The compiler regards an interface as a type.
  - It can be used to declare variables or method parameters

Abstract Classes V.S. Interfaces

- When should one use abstract class instead of interface?
  - If the subclass-superclass relationship is genuinely an "**is a**" relationship.
  - If the abstract class can provide an implementation at the appropriate level of abstraction

- When should one use interface in place of abstract class?
  - When the methods defined represent a small portion of a class
  - When the subclass needs to inherit from another class
  - When you cannot reasonably implement any of the methods

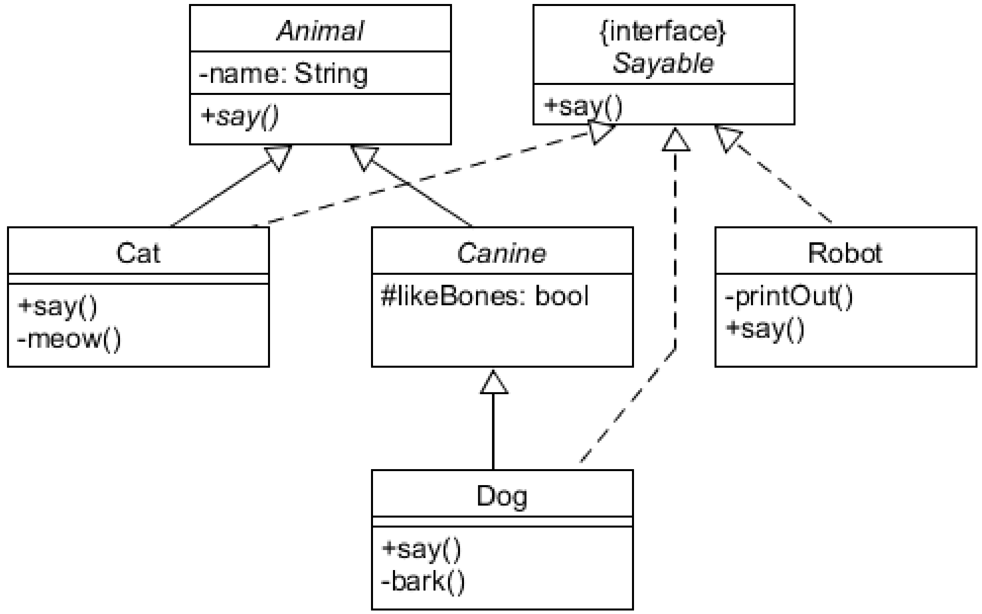
Polymorphism with Interface

```
abstract class Animal {
    protected String name;
    abstract public void say();    }

class Cat extends Animal implements Sayable {
    private void meow() { ... }
    public void say() { meow(); }   }

abstract  class Canine extends Animal {
    protected boolean likeBones;  }

class Dog extends Canine implements Sayable {
    private void bark() { ... }
    public void say() { bark(); }   }

class Robot implements Sayable {
    private void printOut() { ... }
    public void say() { printOut(); }  }

interface Sayable {
    public void say();
}
```

```
Dog   baduki = new Dog();
Cat   nabi = new Cat();
Robot robo = new Robot();

Animal   aref = null;
Sayable sref = null;
Canine cref = null;

aref = baduki; aref.say();
aref = nabi; aref.say();
aref = robo; aref.say();

sref = baduki; sref.say();
sref = nabi; sref.say();
sref = robo; sref.say();

cref = baduki; cref.say();
cref = nabi; cref.say();
cref = robo; cref.say();
```

Polymorphism with Interface

다형성의 핵심은 자식~자손간의 extend 또는 implement 다.




```
Dog   baduki = new Dog();  O
Cat   nabi = new Cat();  O
Robot robo = new Robot();  O

Animal  aref = null;  O
Sayable sref = null;  O
Canine cref = null;  O

aref = baduki; aref.say();  O
aref = nabi; aref.say();  O
aref = robo; aref.say();  X

sref = baduki; sref.say();  O
sref = nabi; sref.say();  O
sref = robo; sref.say();  O

cref = baduki; cref.say();  O
cref = nabi; cref.say();  X
cref = robo; cref.say();  X
```

Polymorphism with Interface

```
abstract class Animal {
    protected String name;
    abstract public void say();    }

class Cat extends Animal implements Sayable {
    private void meow() { ... }
    public void say() { meow(); }    }

abstract  class Canine extends Animal {
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class Dog extends Canine implements Sayable {
    private void bark() { ... }
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class Robot implements Sayable {
    private void printOut() { ... }
    public void say() { printOut(); }   }

interface Sayable {
 public void say(); }
```
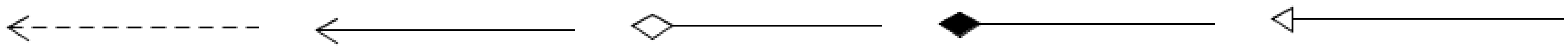

```
Dog   baduki = new Dog();
Cat   nabi = new Cat();
Robot robo = new Robot();

Animal   aref = null;
Sayable sref = null;
Canine cref = null;

aref = baduki; aref.say();
aref = nabi; aref.say();
aref = robo; aref.say();   // Error!

sref = baduki; sref.say();
sref = nabi; sref.say();
sref = robo; sref.say();

cref = baduki; cref.say();
cref = nabi; cref.say(); // Error!
cref = robo; cref.say(); // Error!
```

Class Relationships

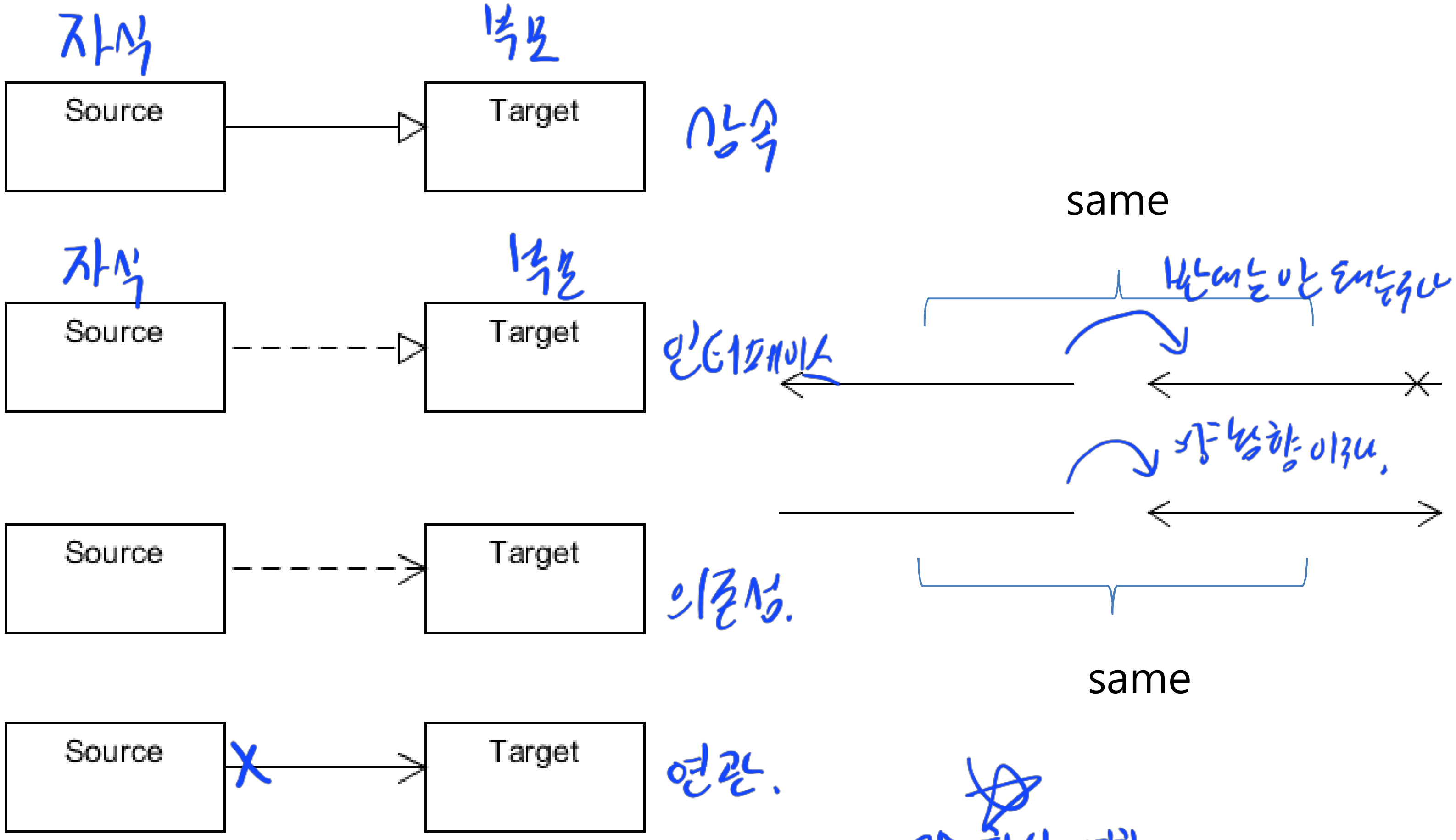


Dependency	Association	Aggregation	Composition	Inheritance
dashed arrow	simple connecting line	empty diamond arrow	filled diamond arrow	empty arrow
when objects of one class work briefly with objects of another class	when objects of one class work with objects of another class for some prolonged amount of time	when one class owns but shares a reference to objects of another class	when one class contains objects of another class	when one class is a type of another class

weaker relationship

stronger relationship

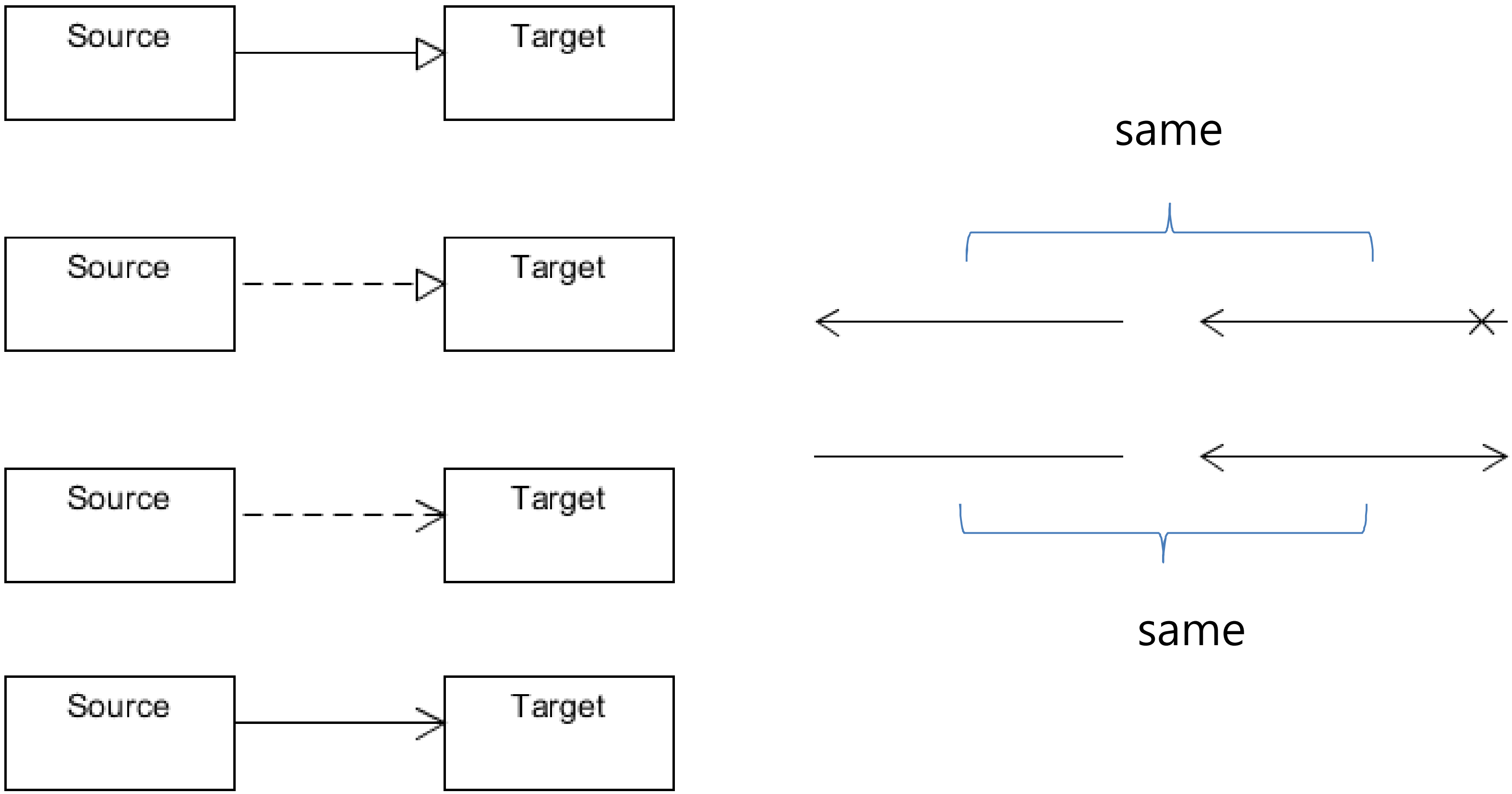
Class Relationships



If Target changes, then Source may be affected.
Target is ignorant of Source.

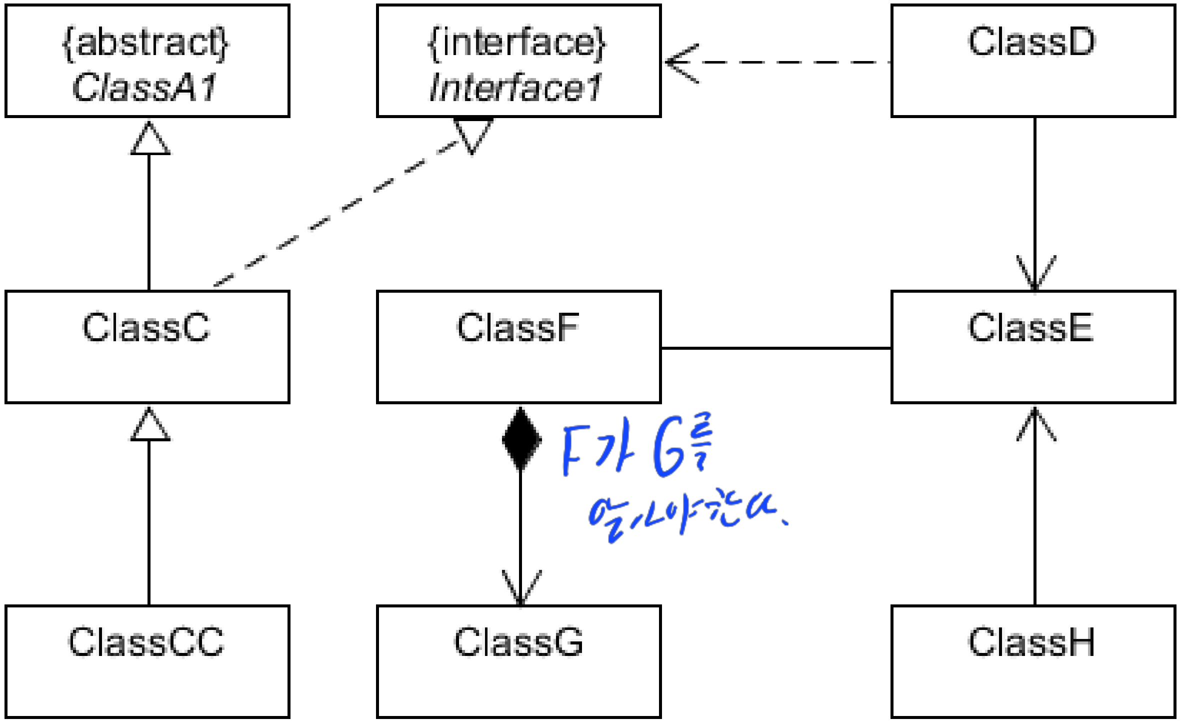
항상 영향 받는건 아님. (단, 잠재적으로 발생가능).

Class Relationships and Change Propagations



If Target changes, then Source may be affected.
Target is ignorant of Source, hence the change of Source is not propagated.

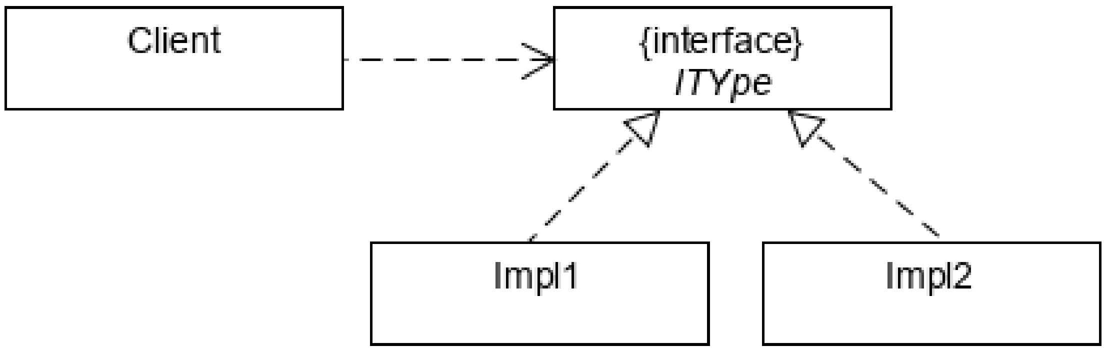
Class Change Propagations



(start) ClassA1 -> ClassC -> ClassCC (end)
(start) ClassD (end)
(start) Interface1 -> ClassC, ClassD -> ClassCC
(start) ClassG -> ClassF -> ClassE -> ClassD, ClassH
(start) ClassE -> ClassD, ClassH, ClassF (end)

Information Hiding

- Design principles for hiding internal design decisions
  - The internal design decisions are most likely to change
  - Thus, reduce the side effects of any future maintenance or modification of the design and hence minimizing the effect on the other modules in the design



Summary

- Object-Oriented Paradigm
  - class, inheritance, polymorphism
- Polymorphism
  - method overloading
  - method overriding
- Class Relationships and Change propagations

OOP Review Quiz : Problem

```
interface IType { }
abstract class Type1 implements IType { }
class Type2 implements IType { }
class Type3 extends Type1 { }
class Type4 extends Type2 { }
```

다음 중 에러 없이 컴파일 되는 코드는?

(a) Type1 a1 = new Type1();

(b) IType a2 = new Type2();

(c) Type1 a3 = new Type2();

(d) IType a4 = new Type4();

(e) Type2 a5 = new IType();

(f) Type1 a6 = new Type3();

(g) Type3 a7 = new Type4();

OOP Review Quiz : Answer

```
interface IType { }
abstract class Type1 implements IType { }
class Type2 implements IType { }
class Type3 extends Type1 { }
class Type4 extends Type2 { }
```

다음 중 에러 없이 컴파일 되는 코드는?

(a) Type1 a1 = new Type1();

(b) IType a2 = new Type2();

(c) Type1 a3 = new Type2();

(d) IType a4 = new Type4();

(e) Type2 a5 = new IType();

(f) Type1 a6 = new Type3();

(g) Type3 a7 = new Type4();